Contact

Fort Collins, Colorado
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Top Skills

Sales Analysis
PivotTables
Inventory Planning

Languages

English (Professional Working)
Hindi (Native or Bilingual)
German (Elementary)

Certifications

Complete A.I. & Machine Learning, Data Science Bootcamp
Microsoft Excel: Advanced Excel Formulas & Functions
Google Data Analytics Professional Certificate
Introduction to Python
The Complete SQL Bootcamp: Go from Zero to Hero

Honors-Awards

Leadership Excellence Award
50% Merit-Based Tuition Scholarship – Eight Consecutive Semesters
Silver Medal – Second Rank in Academic Performance
Appreciation Award – TELNET International Conference
Academic Excellence Award – First Rank in Telecommunications Subjects

Anmol Tripathi

Data Scientist | Machine Learning Engineer | Applied AI, NLP, RAG & LLMs | Transformers, Predictive Modeling | Python, SQL
Fort Collins, Colorado, United States

Summary

I'm a Data Scientist and Machine Learning Engineer with 7+ years of experience across network operations, analytics, predictive modeling, deep learning, NLP, and Applied AI. I build reliable systems that transform complex documents, structured data, and time-series information into grounded answers, defensible predictions, and decision-ready insights.At Hach, I develop AI and machine-learning solutions for product-quality intelligence. My current work includes an internal RAG and LLM-powered agent that retrieves source-grounded information from structured GCS records dating back to 2015 and thousands of NCN documents. It helps R&D and quality teams investigate historical product issues, retrieve supporting evidence, and preserve institutional knowledge across databases and document repositories.I also developed a multi-stage NLP framework for predicting interconnected quality categories—including Failure Mode, Defect Symptom, and Root Cause Function. The hybrid architecture combines Transformer representations, word- and character-level sparse NLP, structured LightGBM models, probability calibration, ensemble learning, and chronological validation. Alongside model development, I automate recurring quality analytics, root-cause investigations, KPI reporting, and executive-ready visualization.Previously, at the University of Arizona College of Nursing, I developed and evaluated LSTM, Bidirectional LSTM, CNN, and Autoencoder architectures for longitudinal maternal-temperature data and labor-timing research. My earlier progression at Orange Business Services established my foundation in Python, SQL, network analytics, anomaly investigation, time-series analysis, data visualization, and operational automation.My public AI/ML portfolio demonstrates selected end-to-end work across Transformer fine-tuning, neural retrieval and reranking, instruction tuning with LoRA and PEFT, encoder-decoder systems, computer vision, forecasting, statistical learning, and predictive modeling. Flagship projects include reproducible code, documented evaluation, model cards, automated validation, and

interactive demonstrations deployed through Hugging Face, Vercel, Streamlit, and GitHub Pages.I’m best aligned with Data Scientist, Machine Learning Engineer, NLP, and Applied AI opportunities where rigorous evaluation, practical implementation, and clear technical decisions matter as much as model sophistication.

Experience

Hach

Quality Data Scientist
September 2024 - Present (2 years)
United States

1) Designed and deployed an internal AI-powered quality intelligence agent using RAG and LLMs, making source-grounded product-quality knowledge accessible to approximately 50 potential users across R&D and quality teams.

2) Integrated the agent with GCS records dating back to 2015 and thousands of NCN PDFs, implementing ingestion, chunking, embeddings, semantic retrieval, contextual prompting, and evidence-based response generation.

3) Enabled product-specific and historical quality questions through a conversational interface, with typical responses generated in approximately 10–15 seconds. This reduces manual searches across GCS, spreadsheets, and documents for supported questions; larger data requests require additional processing time.

4) Grounded responses in retrieved GCS records and NCN documents, allowing users to trace answers to historical evidence instead of relying solely on unconstrained LLM output.

5) Developed a multi-stage NLP framework for predicting Failure Mode, Defect Symptom, and Root Cause Function, using chronological validation to estimate performance on future operational data.

6) Built a calibrated hybrid architecture combining Transformer representations, sparse word- and character-level NLP, structured LightGBM models, probability calibration, and ensemble prediction. In the latest internal full-row Defect Symptom evaluation, external accuracy improved from 69.3%

to 70.1%, macro F1 from 56.1% to 57.7%, and high-confidence auto-accept coverage from 38.4% to 46.0%.

7) Automated monthly, biweekly, and rolling-period quality analytics using Python, SQL Server, Power BI, and Excel, covering product deltas, KPI monitoring, root-cause investigation, data validation, Chemkey testing, and executive reporting while collaborating with quality, engineering, R&D, operations, and business teams.

The University of Arizona College of Nursing
Machine Learning Engineer
May 2023 - August 2024 (1 year 4 months)

Tucson, AZ

1) Led the development of deep-learning and predictive-analytics workflows using longitudinal wearable-ring temperature data from approximately 135 research participants to estimate labor timing and investigate maternal temperature patterns.

2) Extracted per-second observations and minute-level aggregations from SQL Server, then performed data cleaning, statistical analysis, normalization, temporal feature engineering, sequence generation, and ARIMA/SARIMA-based trend analysis to characterize circadian and longitudinal patterns.

3) Designed, trained, tuned, and compared Support Vector Machine, LSTM, Bidirectional LSTM, CNN, and Autoencoder approaches against an existing neural-network baseline, using internal validation and prediction-error analysis to evaluate competing architectures.

4) Identified LSTM-based sequence modeling as the strongest approach among the evaluated architectures and refined the estimated labor-prediction window from approximately 14 days to approximately one day during internal research-model evaluations.

5) Built the end-to-end experimental workflow covering SQL extraction, preprocessing, model development, hyperparameter tuning, validation, architecture comparison, error analysis, and iterative refinement; communicated findings and limitations to interdisciplinary collaborators without presenting research estimates as clinically validated predictions.

University of Arizona BookStores

1 year

Student Assistant Manager

December 2022 - August 2023 (9 months)

Tucson, AZ

1) Promoted from Student Assistant to Student Assistant Manager within four months, combining retail operations, team leadership, and applied analysis of sales, inventory, customer, and marketing data.

2) Analyzed approximately 80,000–100,000 monthly sales records across 30+ product categories using Excel, identifying demand patterns, inventory movement, replenishment requirements, and stock risks.

3) Supported inventory planning and demand forecasting through historical sales analysis, contributing to more informed replenishment decisions and an approximately 15% reduction in overstock and stock-shortage conditions.

4) Developed weekly reports and Tableau dashboards covering sales performance, inventory movement, customer behavior, and category-level trends to support operational, marketing, and product-positioning decisions.

5) Trained and mentored approximately 8–10 team members on operational procedures, reporting practices, data-tracking processes, and analytical tools.

Student Assistant

September 2022 - December 2022 (4 months)

Tucson, AZ

1) Analyzed customer, sales, and purchase-pattern data using Excel to identify trends supporting marketing, promotional, and store-level decisions.

2) Cleaned and validated recurring retail datasets using formulas, filters, lookup functions, and structured worksheets, maintaining accurate sales and customer reporting.

3) Prepared weekly visual reports covering sales performance, customer behavior, and product-level trends while supporting daily operations—establishing the foundation for promotion to Student Assistant Manager within four months.

Orange Business Services

3 years 3 months

Network Operations Center Engineer

April 2022 - August 2022 (5 months)

India

1) Analyzed network-performance, incident, and equipment data using Python and SQL to identify anomalies, recurring fault patterns, service risks, and opportunities for proactive intervention across Cisco and Juniper environments.

2) Developed and evaluated a predictive machine-learning proof of concept for network-failure detection, achieving approximately 85% accuracy during internal model evaluation.

3) Automated recurring data extraction, validation, fault analysis, and performance reporting, improving the consistency of operational investigations and reducing repetitive manual effort.

4) Enhanced Tableau dashboards for 5+ critical network KPIs, enabling stakeholders to monitor network health, incident trends, equipment availability, and operational performance.

5) Performed network-impact analysis, root-cause investigation, incident escalation, carrier coordination, and remote technical support while translating analytical findings into actionable recommendations.

Associate NOC Engineer

December 2019 - March 2022 (2 years 4 months)

Gurugram

1) Monitored enterprise network infrastructure and analyzed more than 10,000 network-performance metrics daily, identifying anomalies, service degradation, recurring faults, and emerging reliability risks across Cisco and Juniper environments.

2) Used Python, SQL, and Excel to extract, clean, validate, and analyze network-performance, incident, and equipment data for proactive monitoring, fault investigation, and operational reporting.

3) Wrote SQL queries to examine historical incidents and performance trends, supporting fault correlation, recurring-problem identification, and root-cause analysis.

4) Created Tableau dashboards and visual reports covering critical network KPIs, equipment availability, incident patterns, and operational performance.

5) Supported end-to-end incident management and equipment operations through network troubleshooting, escalation, carrier coordination, RMA tracking, replacement planning, and restoration documentation.

Graduate Engineering Trainee

June 2019 - December 2019 (7 months)

Noida

1) Completed structured training in enterprise network operations, incident management, monitoring, and troubleshooting across Cisco and Juniper routing and switching environments.

2) Monitored network alarms, performance indicators, device logs, and service events, assisting with issue identification, documentation, escalation, and resolution.

3) Developed foundational capabilities in Python, SQL, Excel, data analysis, and operational reporting to organize network information and identify recurring performance patterns.

4) Assisted senior engineers with troubleshooting, carrier coordination, service restoration, technical documentation, and remote infrastructure support—building the foundation for progression into the Associate NOC Engineer role.

Education

University of Arizona

Masters , Data Science · (August 2022 - December 2023)

Texas McCombs School of Business

Post Graduation Program, Data Science and Business Analytics · (August 2021 - September 2022)

Amity University

Bachelor's degree, Electronics and Telecommunications · (August 2015 - May 2019)

KDMA International

Senior Secondary Education (Class XII), Science – Physics, Chemistry and Mathematics (PCM) · (April 2014 - March 2015)

Puranchandra Vidyaniketan

Secondary School Education (Class X), General Studies · (April 2012 - March 2013)